ExpNo 18 Experiment to understand KNN algorithm for a given data set

Program:

```
import numpy as np

import pandas as pd

df=pd.read_csv('Iris.csv')

df.info()

df.variety.value_counts()

df.head()

features=df.iloc[:,:-1].values

label=df.iloc[:,4].values

from sklearn.model_selection import train_test_split

from sklearn.neighbors import KNeighborsClassifier

xtrain,xtest,ytrain,ytest=train_test_split(features,label,test_size=.2,random
_state=1)

model_KNN=KNeighborsClassifier(n_neighbors=5)

model_KNN.fit(xtrain,ytrain)

print(model_KNN.score(xtrain,ytrain))

print(model_KNN.score(xtest,ytest))

from sklearn.metrics import confusion_matrix

confusion_matrix(label,model_KNN.predict(features))

from sklearn.metrics import classification_report

print(classification_report(label,model_KNN.predict(features)))
```

Output:

```
<class 'pandas.DataFrame'>
RangeIndex: 30 entries, 0 to 29
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   Id            30 non-null     int64
 1   SepalLengthCm  30 non-null    float64
 2   SepalWidthCm   30 non-null    float64
 3   PetalLengthCm  30 non-null    float64
 4   variety       30 non-null     str
dtypes: float64(3), int64(1), str(1)
memory usage: 1.3 KB
1.0
1.0
              precision    recall  f1-score   support

      Setosa       1.00      1.00      1.00        10
  Versicolor       1.00      1.00      1.00        10
   Virginica       1.00      1.00      1.00        10

    accuracy                           1.00        30
   macro avg       1.00      1.00      1.00        30
weighted avg       1.00      1.00      1.00        30
```